upGrad

Summary

Case Study: RFM Models

In this session, we built a dashboard in Tableau to categorise the customers into different segments based on their purchase patterns and spending habits. These categories were based on the recency, frequency and monetary data of the customers.
We also saw how nearly 48% of the revenue is generated by only 10% of the customers, which allows the company to prioritise its marketing efforts accordingly.

Understanding the Business Case

The RFM (Recency, Frequency, Monetary) model is used for fast-moving industries, such as e-commerce and FMCG. The executives responsible for the success of the brand can monitor the consumption patterns of their customers through the lens of their recency, frequency and monetary spending habits.

This case pertains to an online distributor who wants to categorise his customer base into smaller segments. The distributor can neither have a separate service strategy for each customer nor use a one-size-fits-all approach.

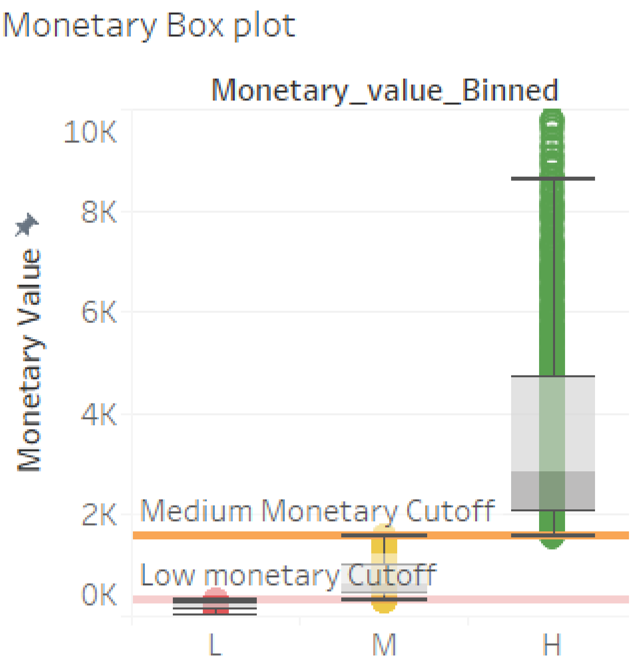
Thus, the distributor plans to create segments within his customer base based on customer purchase patterns.
The customers are segmented into high, medium and low for each category of the Recency, Frequency and Monetary parameters.

You can find the customer data attached here, with the **Recency, Frequency and Monetary** tags given in the session for each customer.

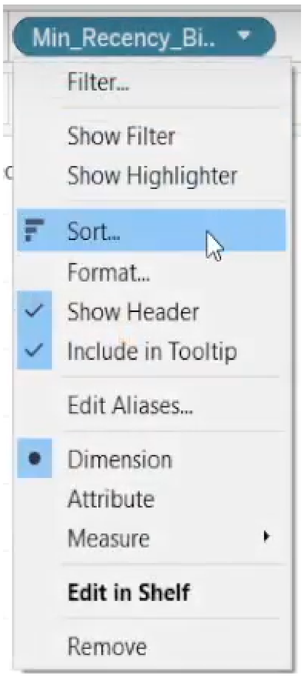
You can access the dashboard used in this session here: Link
The dashboard used in this session can be downloaded from here: Link

Data Quartiles

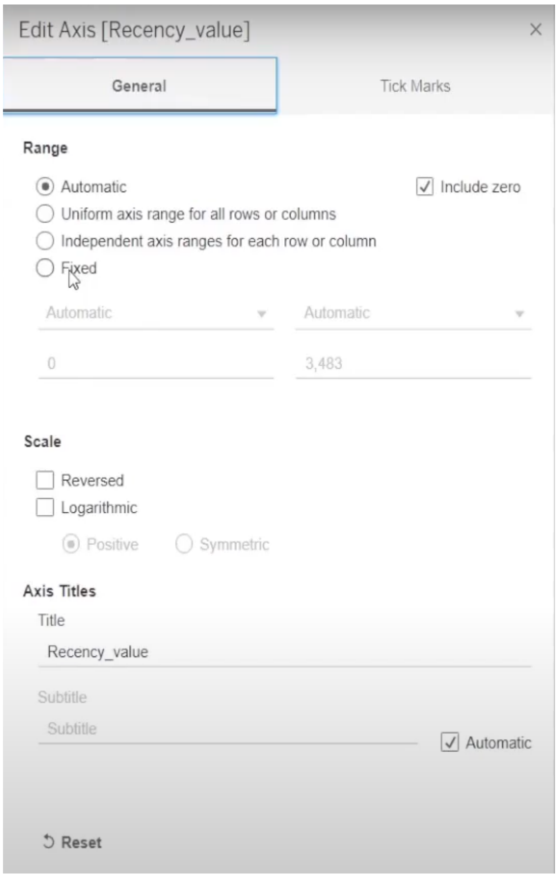
Each category in the RFM analysis is further divided into high, medium and low.
High, medium and low are again divided into different categories based on the top, medium and bottom quartile data.
High values have a disproportionate number of values.
This corresponds to the 80-20 rule that you learnt about earlier, which states that 80% of the revenues for a company are generated by 20% of the customers.



In this session, we saw that Tableau implements a general formatting to the data, but it is not necessarily useful to the business scenario.
We can edit the formatting applied on a variable by selecting the Sort and Format options from the drop-down menu for the variables.



We also understood how we can improve the visibility of charts by editing the values on the axis.



This session also covered the concept of **Aggregate Measure,** which aggregates all the values in a column to a single value. This value can either be the sum, the average or any other value, as required by the business case.

Key Metrics in Data

We can create a table containing the in-depth analytical information about each of the Recency, Frequency and Monetary tags by dragging the **Measure Names** (blue box) to the Rows section of the worksheet and the **Measure Values** (green box) to the Text card under the **Marks** section.

Measure values are not a part of the original dataset connected to Tableau. These values are created by Tableau automatically and contain all possible calculations that can be performed with the existing dataset.

It contains metrics such as:

1. Average of all values,
2. Sum of all values,
3. Maximum values, and
4. Minimum values.

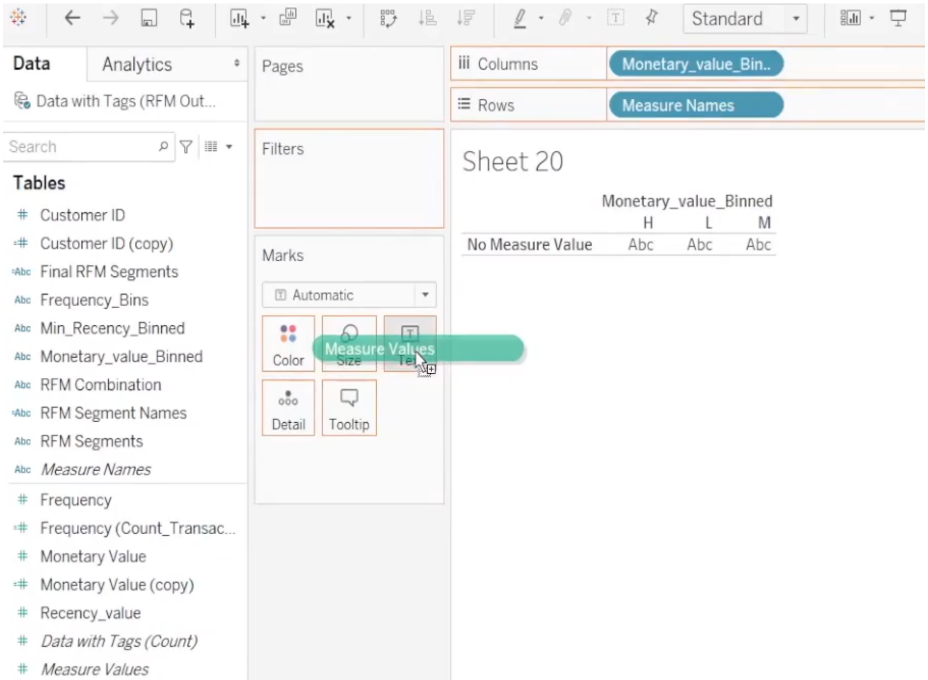
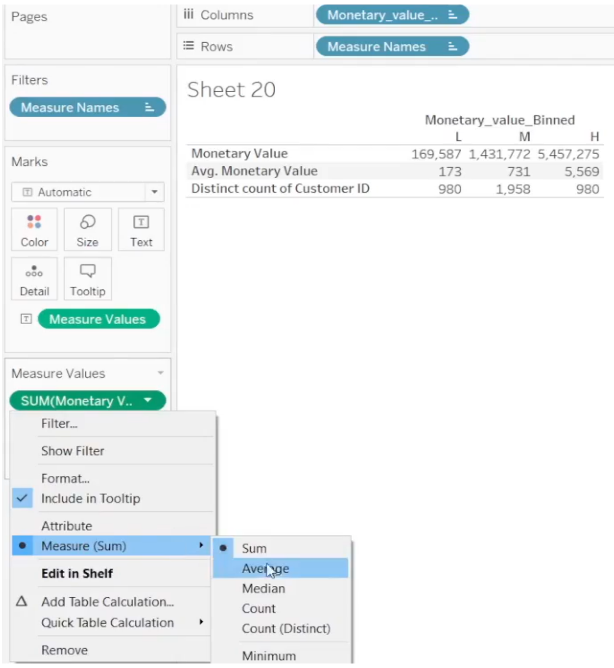


Tableau automatically assigns all possible Measure Values on its own, which can be removed by dragging those values out of the **Measure Values** card.

We can add additional values in the **Measure Values** card by dragging the data onto the card and selecting the desired metric from the drop-down menu for the variable.



Identifying Customer Segments

count by 27 subgroups

Min_Rec..	Frequency_B..	Monetary_value_Binned H	M	L
H	H	406	96	1
	M	92	260	34
	L	3	33	56
M	H	282	152	
	M	136	791	140
	L	22	153	282
L	H	21	22	
	M	12	323	123
	L	6	128	344

unt by 27 subgroups%

in_Rec..	Frequency..	Monetary_value_Binned H	M	L
H	H	10.36%	2.45%	0.03%
	M	2.35%	6.64%	0.87%
	L	0.08%	0.84%	1.43%
M	H	7.20%	3.88%	
	M	3.47%	20.19%	3.57%
	L	0.56%	3.91%	7.20%
L	H	0.54%	0.56%	
	M	0.31%	8.24%	3.14%
	L	0.15%	3.27%	8.78%

revenue for 27 subgroups

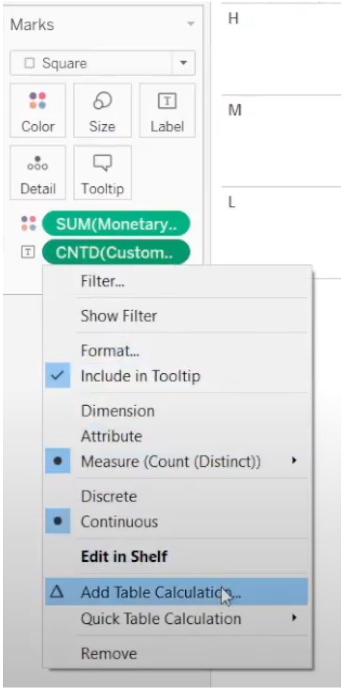
Min_Rec..	Frequency_B..	Monetary_value_Binned H	M	L
H	H	$3,344.72K	$97.36K	$0.27K
	M	$391.78K	$210.83K	$7.19K
	L	$18.03K	$22.07K	$9.46K
M	H	$1,049.89K	$148.71K	
	M	$406.70K	$594.74K	$28.15K
	L	$60.38K	$86.56K	$47.02K
L	H	$57.75K	$19.69K	
	M	$35.74K	$191.59K	$25.25K
	L	$92.28K	$60.23K	$52.24K

revenue for 27 subgroups %

Min_Rec..	Frequency..	Monetary_value_Binned H	M	L
H	H	47.38%	1.38%	0.00%
	M	5.55%	2.99%	0.10%
	L	0.26%	0.31%	0.13%
M	H	14.87%	2.11%	
	M	5.76%	8.43%	0.40%
	L	0.86%	1.23%	0.67%
L	H	0.82%	0.28%	
	M	0.51%	2.71%	0.36%
	L	1.31%	0.85%	0.74%

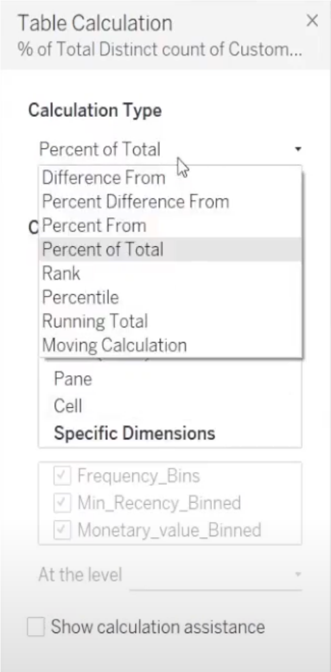
We take the binned values for each variable to the rows and columns of a new worksheet to create a cross-tabulated view for the Recency, Frequency and monetary values.
After formatting the table values to suit our needs, we identify the value that we need to calculate.

These values can be seen in the new worksheet that we have created by dragging the variables to the Text card under Marks. We can add formatting to the data on the basis of the monetary value of each cell by selecting the Colored Table option under the Show Me button.
We can change the reference of the colour code by clicking on the Edit in Shelf option in the drop-down menu of the Color option.
To present data as a percentage of the total data, we need to click on the drop-down menu of the variable and select Table Calculation.



In the Table Calculation menu, we need to select the Percent of Total option as the Calculation option.

A field with a table calculation added to it has a small triangle drawn next to the variable in the Marks card.



It is evident from this dashboard that only ~10% of the entire customer base lies in the HHH zone and is responsible for ~47% of the total revenue.

Thus, these customers are of utmost importance to the company and should be given special attention. In the next segment, we will look at the remaining customers and group them together in a logical manner.

Building Customer Segments

RFM gives us 27 categories based on the purchase patterns and spending habits of the customers.
It is not possible for any company to serve the needs of 27 different customer categories at the same time.

Many categories within the RFM segments within this case have ~0.5% of the total customer base. Most of these segments have a negligible contribution to the total revenue.
Thus, it is best suited to combine multiple segments into one category, as it allows the company to use its resources better.

The upper management has decided to split the customers into seven different segments based on the Recency, Frequency and Monetary variables for a customer.

		% of Revenues	% of Customers	Segment Names
Segment 1	HHH	47%	10%	High Value Loyal Customers
Segment 2	MHH	15%	7%	High Value Loyal Customers (at Risk)
Segment 3	Medium Recency. Any Frequency, Medium-Low Monetary	13%	39%	Passive customers
Segment 4	Medium+High Recency, Low - Medium Frequency, But High Monetary	12%	6%	High Value, Need to be nurtured customers
Segment 5	All Low Recency	5%	24%	Medium-low value, Lapsed/Dormant Customers
Segment 6	High Recency, Any Frequency, Medium-Low Monetary	5%	12%	Exploring customers
Segment 7	High Monetary but Low Recency, any Frequency customers	3%	1%	High Value, Lapsed Dormant customers

The logic used to build the customer segments can be seen here: [Link]
This logic was implemented in Tableau using a **Calculated Field.**
The conditions for the seven segments specified by the upper management were implemented using If-Else-If statements.

The Calculated Field was then used to divide the customers into seven segments and find the division of the number of customers and the revenues for each segment.

Conclusion From the Analysis

Each segment of customers can be treated differently by the organisation with respect to the revenue and the size of the segment.

It is quite obvious that Segment 1 is the most important segment of the entire customer base, as it brings in 47% of the total revenue.
Other segments that can be explored are as follows:

1. Segment 3
    a. This segment contains the highest number of customers.
    b. By interacting with the customers in this segment, we can determine the steps required to convert these customers to higher-paying brackets.
2. Segment 7
    a. These customers are dormant or lapsed customers.
    b. The company can ignore these customers completely, without having any impact on the revenue whatsoever.